SOLUTIONS

illuminati - 2019

Advanced Mathematics Assignment-1B | Algebra

Section 1	Single Correct Answer Type

1.(B) $S_n = 2\left[\frac{1}{2} - \frac{1.3.5......(2n-1)(2n+1)}{2.4.6..........(2n)(2n+2)}\right] \Rightarrow S_\infty = 1$

2.(A)

3.(D) $t_{r+1} = rS_r = r\left[t_r + S_{r-1}\right] = r^2 S_{r-1} = r^2(r-1)S_{r-2}$

$\Rightarrow \quad t_{r+1} = r(r!)t_1 \quad (\because S_1 = t_1)$

$\Rightarrow \quad S_r = (r!)t_1$

$t_{r+1} + S_r = (r+1)S_{r+1} - (r+1)t_{r+1}$

4. (A) $\sum_{r=1}^{n}\left[(r+1)^{P+1} - r^{P+1}\right] = (P+1)\sum_{r=1}^{n} r^P + {}^{P+1}C_2 \sum_{r=1}^{n} r^{P-1} + ... + \sum_{r=1}^{n}(1)$

$a_0 = \frac{1}{P+1}$

5.(C) Given that $f(x) = |a^2x^2 + b|x| + c|$

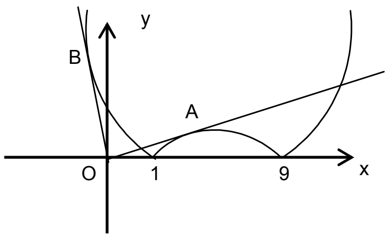
$\Rightarrow \quad f'(x) = \frac{a^2x^2 + b|x| + c}{|a^2x^2 + b|x| + c|}\left(2a^2x + \frac{bx}{|x|}\right)$

$\Rightarrow \quad |a^2x^2 + b|x| + c|$ will have two roots.

Since $|a^2x^2 + b|x| + c|$ is an even function, so the roots will be numerically equal

$\Rightarrow \quad$ sum of the roots = 0.

6.(A) For the given condition to be satisfied the vale of k must lie between the slopes of line OA and OB for slope of line OA



$-(x^2 - 10x + 9) = kx$ must have equal roots

$\Rightarrow \quad (k-10)^2 - 36 = 0 \quad \Rightarrow \quad (k-10)^2 - 36 = 0$

$\Rightarrow \quad k = 4, 16 \quad \Rightarrow \quad k = 4$

For slope of OB

$x^2 - 10x + 9 = k$ must have equal roots

$\Rightarrow \quad (k+10)^2 - 36 = 0 \quad \Rightarrow \quad k = -4, -16$

$\Rightarrow \quad k = -16$

$\Rightarrow \quad k$ should lie between $[4, \infty) \cup (-\infty, -16]$.

7.(D) Line $ax + by + c = 0$ lies in two quadrants

$\Rightarrow$ $abc = 0$; but a and b can't be zero simultaneously also a and c or b and c cannot be zero simultaneously

$\Rightarrow$ $a = 0$ or $b = 0$ or $c = 0$

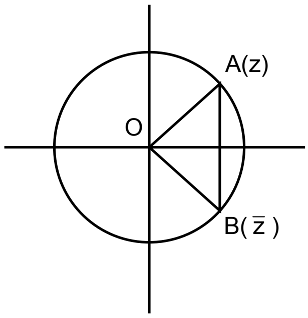
$\Rightarrow$ $D_1 = 0$ or $D_2 = 0$ or $D_3 = 0$

$\Rightarrow$ exactly one of the equations has identical roots.

8.(D) $|z - \overline{z}|$ = straight line AB

while $|z|(\arg z - \arg \overline{Z}) = ArcAB$

$\therefore$ $|z - \overline{Z}| \le |z|(\arg z - \arg \overline{Z})$



9.(C) The given determinant is the product of two determinant as follows.

$$\begin{vmatrix} z & \overline{z} & 1 \\ \overline{z} & z & 1 \\ 1 & z & \overline{z} \end{vmatrix} \times \begin{vmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ a_3 & b_3 & 0 \end{vmatrix} = 0$$

10.(D)

11.(A) Clearly $\angle DOB = \angle COD = A$

$\Rightarrow$ $z = \omega e^{iA}$ and $\overline{\omega} = ze^{iA}$ $\Rightarrow$ $z^2 = \omega\overline{\omega} = 1$

$\Rightarrow$ $z = -1$ (As A and D are on opposite side of BC).

12.(A) $\alpha^n + 2\alpha^{n-1} + 3\alpha^{n-2} - 6 = 18(\alpha - 1)$

$(\alpha^n - 1) + 2(\alpha^{n-1} - 1) + 3(\alpha^{n-2} - 1) = 18(\alpha - 1)$

$= (1 + \alpha + \alpha^2 + \ldots\ldots\ldots + \alpha^{n-1}) + 2(1 + \alpha + \alpha^2 + \ldots\ldots\ldots\ \alpha^{n-2}) + 3(1 + \alpha + \alpha^2 + \ldots\ldots\ldots \alpha^{n-3}) = 18$

$(1 + |\alpha| + |\alpha|^2 + \ldots\ldots |\alpha|^{n-1}) + 2(1 + |\alpha| + |\alpha|^2 + \ldots\ldots |\alpha|^{n-2}) + 3(1 + |\alpha| + |\alpha|^2 + \ldots\ldots\ |\alpha|^{n-3})$

≥ 18

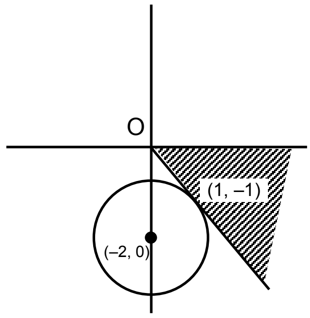
$\frac{1}{1-|\alpha|} + 2\left(\frac{1}{1-|\alpha|}\right) + 3\left(\frac{1}{1-|\alpha|}\right) > 18$ (taking infinite terms and $|\alpha| < 1$)

$\frac{1}{1-|\alpha|} > 3 \quad \Rightarrow \quad |\alpha| > \frac{2}{3}$

13.(B) Distance of z from $(3+2i)$ is same as from imaginary axis $\Rightarrow$ z moves on a parabola with focus $(3+2i)$ and directrix as imaginary axis.

Hence z_1 and z_2 are extremities of focal chord hence $\arg\left(\frac{z_1 - (3+2i)}{z_2 - (3+2i)}\right) = \pm\pi$

14.(C) $\frac{kz}{k+1}$ represent any point lying on the line joining origin and z. So, $\frac{kz}{k+1}$ should lie outside the circle $|z+2| > \sqrt{2}$.



So, z should lie in the shaded region

$\Rightarrow \quad -\frac{\pi}{4} < \arg(z) < 0$

15.(B)

16.(C) A rational number of the desired category is of the form $2014x_1x_2..............x_k$, where $1 \le k \le 9$ and $9 \ge x_1 \ge x_2 \ge ...... \ge x_k \ge 1$. we can choose k digits out of 9 in ${}^9C_k$ ways and arrange them in decreasing order in just one way.

Thus, the desired number of rational number is ${}^9C_1 + {}^9C_2 + ... + {}^9C_9 = 2^9 - 1$

17.(C) Let $t = \frac{2x^2+4}{1+x^2} \quad \Rightarrow \quad 2 < t \le 4$

$\Rightarrow \quad \sin^{-1}(\sin t) < \pi - 3 \quad \Rightarrow \quad \pi - t < \pi - 3 \quad \Rightarrow \quad t > 3$

$\Rightarrow \quad 3 < t \le 4 \quad \Rightarrow \quad |x| < 1$

18.(D) $S_1 + S_2 = \frac{1}{2}\left[\sum_{k=1}^{\infty}\left\{\frac{1}{(6k-1)^2} + \frac{1}{(6k+1)^2}\right\}\right]$

$\Rightarrow \quad 2(S_1+S_2)+1 = \frac{1}{1^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{11^2} + ... \qquad ....(1)$

Let $\quad \alpha = \frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + ... \qquad ....(2)$

$\frac{\alpha}{2^2} = \frac{1}{2^2} + \frac{1}{4^2} + ... \qquad ....(3)$

$\frac{\alpha}{3^2} = \frac{1}{3^2} + \frac{1}{6^2} + ... \qquad ....(4)$

$\frac{\alpha}{6^2} = \frac{1}{6^2} + ... \qquad ....(5)$

Now, (2) – { (4) – (5) + (3)} gives

$\frac{2\alpha}{3} = 2(S_1+S_2)+1$

19.(B) $1 \le \frac{\pi}{\cos^{-1}x} < \infty \quad \Rightarrow 2 \le 2^{\frac{\pi}{\cos^{-1}x}} < \infty$.

Hence 2 should lie between the roots of $t^2 - \left(a + \frac{1}{2}\right)t - a^2 = 0$ where $t = 2^{\frac{\pi}{\cos^{-1}x}}$.

$\Rightarrow \quad a^2 + 2a - 3 \ge 0 \quad \Rightarrow \quad a \in (-\infty, -3] \cup [1, \infty)$

20.(A) Multiplying the given equation by $\frac{c}{a^3}$, we get

$$\frac{b^2c^2}{a^3}x^2 - \frac{b^2c}{a^2}x + c = 0$$

$$\Rightarrow \quad a\left(-\frac{bc}{a^2}x\right)^2 - b\left(\frac{bc}{a^2}\right)x + c = 0 \quad \Rightarrow \quad -\frac{bc}{a^2}x = \alpha, \beta$$

$$\Rightarrow \quad (\alpha+\beta)\alpha\beta x = \alpha, \beta \quad \Rightarrow \quad x = \frac{1}{(\alpha+\beta)\alpha}, \frac{1}{(\alpha+\beta)\beta}.$$

21.(A) α, β are roots of $x^2 - x + q = 0$

$\Rightarrow \quad \alpha + \beta = 1; \alpha\beta = q$

Now $\quad S_{k-1} = S_k - qS_{k-1}$

$S_k = S_{k-1} - qS_{k-2}$

:

:

$S_2 = S_1 - qS_0$

Adding $S_{k-1} = S_1 - q(S_{k-1} + S_{k-2} + ... + S_0) = 1 - q(S_{k-1} + S_{k-2} + ... + S_0)$

22.(C)

$|k + z^2| = |z|^2 - k = |z^2| + |k|$

$\arg(z^2) = \arg(k)$

$\Rightarrow 2\arg(z) = \pi \Rightarrow \arg(z) = \frac{\pi}{2}$.

23.(C) let coefficient of y^m in $e^{xy}\left(e^y - 1\right)^m$

$= $ coefficient of y^m in $e^{xy}\left\{e^{my} - {}^mC_1 e^{(m-1)y} + ... + (-1)^m . {}^mC_m\right\}$

$\Rightarrow \quad$ Coefficient of y^m in $e^{xy}\left\{\left(1 + \frac{y}{1!} + \frac{y^2}{2!} + ...\right) - 1\right\}^m$

$=$ coefficient of y^m in $\left\{e^{xy}\left[e^{my} - {}^mC_1 e^{(m-1)y} + ... + (-1)^m . {}^mC_m\right]\right\}$

$\Rightarrow$ Coefficient of y^m in $e^{xy} y^m \left(1 + \frac{y}{2!} + \frac{y^2}{3!} + ...\right)^m$

$=$ coefficient of y^m in

$\left\{e^{(x+m)y} - {}^mC_1 e^{y(m+x-1)} + ... + (-1)^m \, {}^mC_m . e^{xy}\right\}$

$$\Rightarrow \quad 1 = \frac{(x+m)^m}{m!} - {}^mC_1 . \frac{(x+m-1)^m}{m!} + ... + (-1)^m \frac{{}^mC_m . x^m}{m!}$$

$$\Rightarrow \quad m! = (x+m)^m - {}^mC_1 . (x+m-1)^m$$

$$+ {}^mC_2 (x+m-2)^m + ... + -(-1)^m . {}^mC_m$$

24.(A) If we take boxes to be distinct

Total solution $= {}^{999+3-1}C_2 = 500500$

Total solution (when each box have different number of balls) $= 500500 - 3 \times 499 - 1 = 499002$

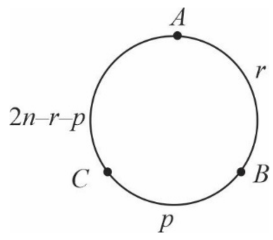
(where 3×499 are the number of solutions when exactly two boxes have same number of balls and there is only one solution when each box have same number of balls).

Now, n such solutions for distinct boxes will correspond to $\frac{n}{6}$ solutions if boxes are identical.

Hence total number of solutions when boxes are identical $= \frac{49002}{6}$ (when each box have different number of balls) + 500 (at least two box have same number of balls)
$= 83667$.

25.(D)

26.(A) Let there are r seats between A and B, p seats between B and C and $2n-r-p$ seats between A and C.



Total number of ways of seating arrangements when there is no restriction $= (2n)!$

Now we count the cases when two particular persons sit together.
Number of ways in which two persons sit together between A and $B = (r-1)2$

Number of ways in which two persons sit together between B and $C = (p-1)2$ and between

A and C these ways are $(2n-r-p-1)2$ so total number of ways of their sitting together

$$= (2n-r-p-1+r-1+p-1)2 = (2n-3)2$$

With each of such cases, remaining persons can sit in $(2n-2)!$ Ways

So total number of cases $= (2n-3)2.(2n-2)!$

The number of ways in which two particular persons does not sit together

$$= (2n)! - 2(2n-3)(2n-2)! = (4n^2 - 6n + 6)(2n-2)!$$

27.(A) $kx + y + z + w = kr$

$0 \le x \le r, \quad 0 \le y \le kr, \quad 0 \le z \le kr, 0 \le w \le kr$

$y + z + w = k(r-x)$

Now taking $x = p, \qquad 0 \le p \le r$

Then $y + z + w = k(r-p)$ has ${}^{k(r-p)+2}C_2$ number of solutions.

Hence total solutions of equation

$$= \sum_{p=0}^{r} {}^{k(r-p)+2}C_2 = \frac{(kr+2-kp)(kr+1-kp)}{2} = \frac{(r+1)}{12}\left(2k^2r^2 + k^2r + 9kr + 12\right).$$

28.(B) The prime digits are 2, 3, 5 and 7. As sum of two digits is also prime, this can be happen with 2, 3 and 2, 5 only. So the $2n$–digit number is comprising only these (2, 3, 5) digits and 2 occurs at every alternate place.
Now,
Case–I:
The number starts with '2'

2		2		2		2		2	

$\uparrow$ 3 or 5 $\quad\uparrow$ 3 or 5 $\quad\uparrow$ 3 or 5 $\quad\uparrow$ 3 or 5 (two ways)

So no of numbers in this case $= 2\times 2\times 2.........n$ times $= 2^n$

Case –II:

The number starts with 3 or 5

	2		2					2

(two ways) 3 or 5 $\quad$ 3 or 5 $\quad$ 3 or 5 $\quad$ 3 or 5 $\quad$ 3 or 5

In this case also, number of numbers $= 2^n$ so total number of such numbers

$= 2\times 2^n = 2^{n+1}$.

29.(A) $\displaystyle\sum_{k=0}^{n+1}\sum_{j=1}^{k+1}\left(\sum_{i=1}^{k} {}^{i+1}C_j - {}^{i}C_j\right) = \sum_{k=0}^{n+1}\sum_{j=1}^{k+1}\left({}^{k+1}C_j - {}^{1}C_j\right) = \sum_{k=0}^{n+1}\left(2^{k+1}-2\right)$

$\displaystyle = \sum_{k=0}^{n+1} 2^{k+1} - 2\sum_{k=0}^{n+1} 1 = 2\cdot\left(2^{n+2}-1\right) - 2\left(n+2\right) = 2^{n+3} - 2n - 6.$

30.(A) ${}^{m}C_n$ is the coefficient of x^n in the binomial expansion of

$\left(1+x\right)^m$. Hence, for $\dfrac{(1+x)^m}{2} < 1,$

$\dfrac{{}^{m}C_n}{2} + \dfrac{{}^{m+1}C_n}{2^2} + \ldots\ldots\ldots$ to $\infty =$ the coefficient of x^n in

$\dfrac{(1+x)^m}{2} + \dfrac{(1+x)^{m+1}}{2^2} + \dfrac{(1+x)^{m+2}}{2^3} + \ldots\ldots\ldots$ to ∞.

= The coefficient x^n in $\dfrac{\dfrac{(1+x)^m}{2}}{1-\dfrac{1+x}{2}} = \dfrac{(1+x)^m}{1-x}$

= the coefficient of x^n in

$\left(1+x\right)^m \quad (1 + x + x^2 + x^3 + \ldots\ldots\ldots\ldots\ldots$ to $\infty)$

$= {}^{m}C_n + {}^{m}C_{n-1} + {}^{m}C_{n-2} + \ldots\ldots + {}^{m}C_2 + {}^{m}C_1 + {}^{m}C_0.$

Section 2	**Link Comprehension Type**

31-32. 31.(B) 32.(C) 33.(D)

31.(B) We have, $\left(1+px+x^2\right)^n = 1 + a_1x + a_2x^2 + \ldots\ldots\ldots\ldots + a_{2n}x^{2n}$

Differentiate with respect to x, we get

$n\left(1+px+x^2\right)^{n-1}\left(p+2x\right) = a_1 + 2a_2x + 3a_2x^2 + \ldots.. + 2na_{2n}x^{2n-1}$

Multiply by $\left(1+px+x^2\right)$ both side, we get

$n\left(1+px+x^2\right)^n\left(p+2x\right) = \left(1+pn+x^2\right)\left(a_1 + 2a_2x + 3a_3x^2 + \ldots.. + 2na_{2n}x^{2n-1}\right)$

$n\left(p+2x\right)\left(1+a_2x+a_2x^2+......+a_{2n}x^{2n}\right)=\left(1+px+x^2\right)$

$\left(a_1+2a_2x+3a_3x^2+......+\left(r-1\right)a_{r-1}x^{r-1}+ra_rx^{r-1}+\left(r+1\right)a_{r+1}x^r\right)$

Equating the coefficient of x^r, we get

$nPa_r+2na_{r-1}=\left(r+1\right)a_{r+1}+pra_r+\left(r-1\right)a_{r-1}$

$npa_r-pra_r=\left(r+1\right)a_{r+1}+\left(r-1\right)a_{r-1}-2na_{r-1}$

$\left(np-pr\right)a_r=\left(r+1\right)alr+1+\left(r-1-2n\right)a_{r-1}$

Hence, option **(B)** is correct

32.(C) We have, $\left(1+px+x^2\right)^n=1+a_1x+a_2x^2+a_3x^3+.....+a_{2n}x^{2n}$

On replacing x by x^4, we get

$\left(1+px^4+x^8\right)^n=1+a_1x^4+a_2x^8+a_3x^{12}+......+a_{2n}x^{8n}$

On dividing by x^3 both side, we get

$$\frac{\left(1+px^4+x^8\right)^n}{x^3}=\frac{1}{x^3}+a_1x+a_2x^5+a_3x^9+......+a_{2n}x^{8n-3}$$

[on differentiate w.r.t x, we get]

$$\frac{nx^3\left(4px^3+8x^7\right)\left(1+px^4+x^8\right)^{n-1}-3x^2\left(1+px^4+x^8\right)^n}{x^6}$$

$$=\frac{-3}{x^4}+a_1+5a_2x^4+9a_3x^8+.....+\left(8n-3\right)a_{2n}x^{8n-4}$$

Put $x=1$ both side, we get

$n\left(4p+8\right)\left(2+p\right)^{n-1}-3\left(2+p\right)^n=-3+a_1+5a_2+9a_3+.....+\left(8n-3\right)a_{2n}$

$\left(p+2\right)^n.\left(4n-3\right)+3=a_1+5a_2+9a_3+........+\left(8n-3\right)a_{2n}$

When $a_1+5a_2+9a_3+.......+\left(8n-3\right)a_{2n}$ is divide by $\left(p+2\right)$ the remainder is 3.

Hence, option **(C)** is correct.

33.(D) We have, $\left(1+px+x^2\right)^n=1+a_1x+a_2x^2+a_3x^3+......+a_{2n}x^{2n}$

On replacing x by x^2, we get

$\left(1+px^2+x^4\right)^n=1+a_1x^2+a_2x^4+a_3x^6+......+a_{2n}x^{4n}$

On dividing by x, we get

$$\frac{\left(1+px^2+x^4\right)^n}{x}=\frac{1}{x}+a_1x+a_2x^3+a_3x^5+.....+a_{2n}x^{4n-1}$$

On differentiating w.r.t. to x. we get

$$\frac{nx\left(2px+4x^3\right)\left(1+px^2+x^4\right)^{n-1}-\left(1+px^2+x^4\right)^n}{x^2}$$

$$=\frac{-1}{x^2}+a_1+3a_2x^2+5a_3x^4+....+\left(4n-1\right).x^{4n-2}.a_{2n}$$

Put $p=-3$ and $x=1$

$\Rightarrow\qquad n\left(-6+4\right)\left(1-3+1\right)^{n-1}-\left(1-3+1\right)^n$

$= -1 + a_2 + 3a_2 + 5a_3 + ... + (4n-1)a_{2n}$

$\Rightarrow \quad -2n(-1)^{n-1} - (-1)^n + 1 = a_1 + 3a_2 + 5a_3 + ... + (4n-1)a_{2n}$

$\Rightarrow \quad 2n(-1)^n - (-1)^n + 1 = a_1 + 3a_2 + 5a_3 + ... + (4n-1)a_{2n}$

$\Rightarrow \quad 2n - 1 + 1 = a_1 + 3a_2 + 5a_3 + ... + (4n-1)a_{2n}$

$\Rightarrow \quad a_1 + 3a_2 + 5a_3 + ... + (4n-1)a_{2n} = 2n\left[\because (-1)^n = 1, n \text{ is even}\right]$

Hence Option **(D)** is correct

34-36. 34.(D) 35.(C) 36.(D)

34.(D) Let's count the Δ's right way

FS	**SS**	**TS**
1	1	1
2	2	1, 2, 3
3	3	1, 2, 3, 4
4	4	1, 2, 3, 4

$FS \to$ First side, $SS \to$ Second side

$TS \to$ Third side

$\therefore$ Total number of ways $= 1 + 3 + 4 + 4 = 12$

35.(C) The value of equal sides can vary from 1 to $2P$.

Hence, third side $< 2, < 4, < 6, ..., < 4p$

FS	**SS**	**TS**
1	1	1
2	2	1, 2, 3
3	3	1, 2, 3, 4, 5
4	4	1, 2, 3, 4, 5, 6, 7
:	:	:
p	p	1, 2, …….., (2p-1)
p+1	p+1	1, 2, 3, ……., 2p
:	:	:
2p	2p	1, 2, 3, ….., 2p

$\therefore$ Number of $\Delta's = \{1 + 3 + 5 + 7 + ... + (2p-1)\} + \{2p + 2p + ... + p \text{ times}\} = p^2 + 2p^2 = 3p^2$

36.(D) If $c = 3 \quad \Rightarrow$ triangle having 3 as largest sides and which are isosceles or equilateral are 3, 3, 1 ; 3, 3, 2 ; 3 , 3, 3 ; 3, 2, 2

37-39 37.(C) 38.(B) 39.(A)

$x^2 = z^2 + b^2 - 2bz\cos\alpha$

$y^2 = x^2 + c^2 - 2cx\cos\alpha$

$z^2 = y^2 + a^2 - 2ay\cos\alpha$

adding $2(cx + ay + bz)\cos\alpha = a^2 + b^2 + c^2$

$\Delta = \left(\dfrac{cx + ay + bz}{2}\right)$. $\sin\alpha$, thus $\tan\alpha = \dfrac{4\Delta}{a^2 + b^2 + c^2}$

by sine rule, $x = \dfrac{b\sin\alpha}{\sin A}$, $y = \dfrac{c\sin\alpha}{\sin B}$, $z = \dfrac{a\sin\alpha}{\sin C}$

area of $\Delta CAP = \frac{zx}{2}\angle CPA = \frac{1}{2}\frac{a\sin\alpha}{\sin C}\cdot\frac{b\sin\alpha}{\sin A}\cdot\sin A$

$= \frac{ab\sin C}{2}\cdot\frac{\sin^2\alpha}{\sin^2 C} = \frac{\Delta\sin^2\alpha}{\sin^2 C}$

Similarly area of $\Delta ABP = \Delta\frac{\sin^2\alpha}{\sin^2 A}$

area of $\Delta BCP = \Delta\frac{\sin^2\alpha}{\sin^2 B}$

adding $\text{cosec}^2\alpha = \text{cosec}^2 A + \text{cosec}^2 B + \text{cosec}^2 C$

$\sin^2 A + \sin^2 B + \sin^2 C \le \frac{9}{4}$

$\Rightarrow \quad \text{cosec}^2\alpha \ge 4 \quad \Rightarrow \quad 0 < \alpha \le 30.$

Section 3	Multiple Correct Answer Type

40.(ABC)

Number of all possible triangles $= {}^nC_3$.

Out of these n triangles have two sides common with polygon and

$n(n-4)$ triangles have exactly one side common with polygon.

So, desired number of triangles

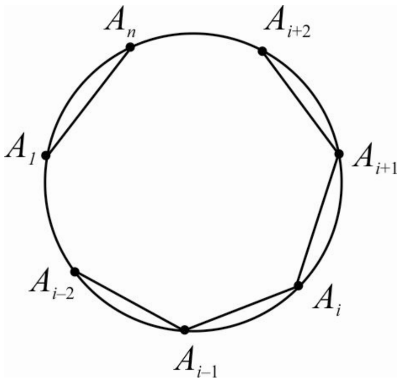
$= {}^nC_3 - n - n(n-4)$

$= \frac{n(n-1)(n-2)}{6} - n - n(n-4)$

$= \frac{n}{6}(n-4)(n-5) = \frac{n}{n-3}\,{}^{n-3}C_3$

ALTERNATIVELY:

If we consider a particular vertex, say A_i. If A_i is not included in the selection then we have to select 3 vertices from remaining $(n-1)$ vertices on a circle such that no two are consecutive, which can be done in ${}^{(n-1)-3+1}C_3$ ways $= = {}^{n-3}C_3$ ways.



If A_i is included, then A_{i-1} and A_{i+1} can not be included, so we can choose 2 points from remaining $n-3$ points such that they are not adjacent in ${}^{n-3-2+1}C_2 = {}^{n-4}C_2$ ways So, desired number of triangles $= {}^{n-3}C_3 = {}^{n-4}C_2$

41.(AB)

The number of ways of selecting r persons from n is ${}^nC_r$. Out of these r persons 3 have to selected for the post of a treasurer, secretary and chair person with repetition. This can be done in $r.\ r.\ r$ ways

Number of ways $= r^3 \cdot {}^nC_r$

Since r can vary from 1 to n, total number of ways $= \sum_{r=1}^{n} r^3\ {}^nC_r$

$$= n\sum_{r=1}^{n} r^2 \cdot {}^{n-1}C_{r-1} = n\sum_{r=1}^{n} r\cdot(r-1+1)\,{}^{n-1}C_{r-1} = n\sum_{r=2}^{n} r\cdot\left[(n-1)\,{}^{n-2}C_{r-2} + {}^{n-1}C_{r-1}\right]$$

$$= n(n-1)\sum_{r=2}^{n}(r-2+2)\,{}^{n-2}C_{r-2} + n\sum_{r=1}^{n}(r-1+1)\,{}^{n-1}C_{r-1}$$

$$= n(n-1)(n-2)\sum_{r=3}^{n} {}^{n-1}C_{r-3} + 2n(n-1)\sum_{r=2}^{n} {}^{n-2}C_{r-2} + n(n-1)\sum_{r=2}^{n} {}^{n-2}C_{r-2} + n\sum_{r=1}^{n} {}^{n-1}C_{r-1}$$

$= n(n-1)\,(n-2).\ 2^{n-3} + 3n\,(n-1).2^{n-2} + n.\ 2^{n-1}$.

Otherwise (Alternative):

The three posts can be allocated as per the cases:

Case I: Same person gets all the three posts.

When a person is having a post then the size of fellow group members in all possible cases will vary from 0 to $n-1$ and every person is equally likely to get all the three posts. So if a person is selected first for the three posts and then the team is constituted this will be equivalent to selecting a team and then giving all the three posts to any one of them (In all possible cases) this can be done in ${}^nC_1\,2^{n-1}$ ways.

Case II: One person gets two posts and one gets one post.

Similarly this can be done in ${}^nC_2\,2^{n-2}\times 3!$ ways.

Case III: All the three posts go to different person.

Similarly this can be done in ${}^nC_3\,2^{n-3}\times 3!$ ways.

Hence the total number of ways $= {}^nC_1\,2^{n-1} + {}^nC_2\,2^{n-2}\times 3! + {}^nC_3\,2^{n-3}\times 3!$

$= 2^{n-1} + 3n(n-1)\,2^{n-2} + n(n-1)(n-2)\,2^{n-3}$.

42.(AB)

$$T_n = \frac{\left(r^2+r-1\right)\cdot r!}{\left(r^2+3r+2\right)} = \frac{\left[\left(r+1\right)^2-\left(r+2\right)\right]\cdot r!}{\left(r^2+3r+2\right)} = \left[\frac{(r+1)\cdot(r+1)! - r!(r+2)}{(r+1)(r+2)}\right]$$

$$T_r = \frac{(r+1)!}{r+2} - \frac{r!}{(r+1)} \quad\Rightarrow\quad T_r = V_r - V_{r-1}$$

$T_1 = V_1 - V_0$

$T_2 = V_2 - V_1$

.

.

.

$T_n = V_n - V_{n-1}$

$S_n = T_1 + T_2 + \ldots.. + T_n$

$S_n = V_n - V_0$

$$S_n = \frac{(n+1)!}{n+2} - \frac{1}{2} \quad\Rightarrow\quad S_n = \frac{2(n+1)! - (n+2)}{2(n+2)}$$

43.(ABD)

44.(AC)

We have ${}^{100}C_6 + 4.{}^{100}C_7 + 6.{}^{100}C_8 + 4.{}^{100}C_9 + {}^{100}C_{10}$

$\Rightarrow \quad {}^4C_0{}^{100}C_{94} + {}^4C_1{}^{100}C_{93} + ... + {}^4C_4{}^{100}C_{90}$

$(1+x)^4 = {}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4$

and $\quad (1+x)^{100} = {}^{100}C_0 + {}^{100}C_1x + ...$

$.............. + {}^{100}C_{94}x^{94} + {}^{100}C_{95}x^{95} + ... + x^{100}$

$(1+x)^4(1+x)^{100} = {}^4C_0 + {}^{100}C_0 + {}^4C_1{}^{100}C_1x^3 + ...$

$(1+x)^{104} = \left[{}^4C_0{}^{100}C_0 + {}^4C_1{}^{100}C_1x^3 + ...\right]$

Coefficient of x^{94}, we get

$${}^{104}C_{94} = {}^4C_0{}^{100}C_{94} + {}^4C_1{}^{100}C_{93} + {}^4C_2{}^{100}C_{92} + {}^4C_3{}^{100}C_{91} + {}^4C_4{}^{100}C_{90}$$

$\therefore \quad x + y = 104 + 94 = 198$

Also, $\quad {}^{100}C_6 + 4.{}^{100}C_7 + 6.{}^{100}C_8 + 4.{}^{100}C_9 + {}^{100}C_{10}$

$\Rightarrow \quad {}^4C_4{}^{100}C_6 + {}^4C_3{}^{100}C_7 + {}^4C_2{}^{100}C_8 + {}^4C_1{}^{100}C_9 + {}^4C_{10}{}^{100}C_{10}$

Coefficient of x^{10} in $(1+x)^{104}$ is

$${}^{104}C_{10} = {}^4C_4{}^{100}C_6 + {}^4C_3{}^{100}C_7 + {}^4C_2{}^{100}C_8 + {}^4C_1{}^{100}C_9 + {}^4C_4{}^{100}C_{10}$$

$\therefore \quad x + y = 104 + 10 = 114$

Hence, Option**(A)** and **(C)**

45.(BC)

We have $\left(x^2 + \frac{1}{x^2} - 2\right)^{10} \Rightarrow \left(x - \frac{1}{x}\right)^{20}$

$T_{r+1} = {}^{20}C_r(x)^{20-r}(-x)^{-r}$

$T_{r+1} = {}^{20}C_r(x)^{20-2r}(-1)^r$

The value of constant term in the binomial expansion $\left(x - \frac{1}{x}\right)^{10}$, if

$20 - 2r = 0 \Rightarrow r = 10$

$T_{10+1} = {}^{20}C_{10}(-1)^{10} = {}^{20}C_{10}$

(A) Number of different dissimilar term in the expansion

$(x_1 + x_2 + x_3 + ... + x_{10})^{10} = {}^{10+10-1}C_{10-1} = {}^{19}C_9$

(B) $\left({}^{10}C_0\right)^2 + \left({}^{10}C_1\right)^2 + ... + \left({}^{10}C_{10}\right)^2 = \frac{(2\times10)!}{10!10!} = \frac{20!}{10!10!} = {}^{20}C_{10}$

$\left[\because \left({}^nC_1\right)^2 + \left({}^nC_2\right)^2 + ... + \left({}^nC_n\right)^2 = \frac{2n!}{n!n!}\right]$

(C) Coefficient of x^{10}, in $(1-x)^{20} = {}^{20}C_{10}(-1)^{10} = {}^{20}C_{10}$

(D) Number of linear arrangement of 20 things of which 10 alike of one kind and rest all are different, taken all at a time $= \frac{20!}{10!}$

46.(BCD)

We know that, distribution of n distinct objects in to r different boxes, if empty boxes are not allowed or in each box atleast one object is put $(n > r)$.

The number of ways

$$= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n + ...(-1)^{r-1}\,{}^nC_{r-1}1 \qquad ....(i)$$

Hence, $n = 5, r = 3$

$\therefore$ number of ways in which the different books be distributed in 3 persons, so each gets atleast one is $(3)^5 - {}^3C_1(2)^5 + {}^3C_2 = 243 - 3 \times 32 + 3 = 150$

(B) Number of parallelogram formed by one set of 6 parallel line and other set of 5 parallel lines $= {}^6C_2 \times {}^5C_2 = 15 \times 10 = 150$

(C) 5 different toys to be distributed among 3 children, so each gets atleast one $= (3)^5 - {}^3C_1(2)^5 + {}^3C_2 = 243 - 3 \times 32 + 3 = 150$

(D) 3 mathematics professors are assigned five different lectures, so that each professor get at least one lecture.

$= (3)^5 - {}^3C_1(2)^5 + {}^3C_2 = 243 - 3 \times 32 + 3 = 150$

47.(BCD)

Let $\theta = 2\tan^{-1}(-3) \Rightarrow -\pi < \theta < 0$

$$\tan\frac{\theta}{2} = -3 \Rightarrow \tan\theta = \frac{2\tan(\theta/2)}{1-\tan^2(\theta/2)} = \frac{3}{4} \Rightarrow -\pi < \theta < -\frac{\pi}{2}$$

Let $\alpha = \tan^{-1}\left(\frac{3}{4}\right)$ (principal value) then

$\theta = \alpha - \pi$

$$\text{now } \cos\alpha = \frac{4}{5} \Rightarrow \theta = \cos^{-1}\left(\frac{4}{5}\right) - \pi = \frac{\pi}{2} - \sin^{-1}\left(\frac{4}{5}\right) - \pi = -\sin^{-1}\left(\frac{4}{5}\right) - \frac{\pi}{2}$$

$$\cot\alpha = \frac{4}{3} \Rightarrow \theta = \cot^{-1}\left(\frac{4}{3}\right) - \pi = \frac{\pi}{2} - \tan^{-1}\left(\frac{4}{3}\right) - \pi = -\tan^{-1}\left(\frac{4}{3}\right) - \frac{\pi}{2}.$$

48.(BCD)

$$\tan^{-1}\frac{a}{x} + \tan^{-1}\frac{b}{x} = \frac{\pi}{2} - \left(\tan^{-1}\frac{c}{x} + \tan^{-1}\frac{d}{x}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{a}{x}+\frac{b}{x}}{1-\frac{ab}{x^2}}\right) = \frac{\pi}{2} - \tan^{-1}\left(\frac{\frac{c}{x}+\frac{d}{x}}{1-\frac{cd}{x^2}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{(a+b)x}{x^2-ab}\right) = \cot^{-1}\left(\frac{(c+d)x}{x^2-cd}\right) \Rightarrow \left(\frac{(a+b)x}{x^2-ab}\right) = \left(\frac{x^2-cd}{(c+d)x}\right)$$

$$\Rightarrow x^4 - (ab+ac+ad+bc+bd+cd)x^2 + abcd = 0$$

its roots are x_1, x_2, x_3, x_4

$\therefore \sum x_1 = 0$, $\sum x_1x_2 = -\sum ab$, $\sum x_1x_2x_3 = 0$ and $x_1x_2x_3x_4 = abcd$

$\therefore \sum \frac{1}{x_1} = 0$ and $(x_2+x_3+x_4)(x_3+x_4+x_1)(x_4+x_1+x_2)(x_1+x_2+x_3) = x_1x_2x_3x_4$

49.(ABC)

$\sin^{-1}(a^2x^2 + b^2y^2) + \cos^{-1}|ax + by| = \pi$

$\Rightarrow \quad a^2x^2 + b^2y^2 = 1$ and $|ax + by| = 0 \quad \Rightarrow \quad -2abxy = 1$

50.(ABD)

$t_2 = t_1 + t_3$

$t_3 = t_2 + t_4$

$\vdots$

$\vdots$

$t_9 = t_8 + t_{10}$

$t_1 + (t_3 + t_4 + t_5 + ... + t_8) + t_{10} = 0$

51.(AB)

x can not be odd integer for if x is odd, x^2 is odd but $2px + 2q$ is even;

so $x^2 + 2px + 2q \neq 0$

x can not be even integer for if x is even, $x^2 + 2px$ is a multiple of 4 but $2q$ is not.

So $x^2 + 2px + 2q \neq 0$

Also $(x + p)^2 = p^2 - 2q$

$\Rightarrow$ If x is fraction then $(x + p)^2$ is also a fraction but $p^2 - 2q$ is an integer. So, roots cannot be integer or rational numbers.

52.(ABC)

Let $f(x) = ax^2 + bx + c$

-3 and 3 are lying between the roots of $f(x) = 0$

$\Rightarrow \quad f(-3) > 0$ and $f(3) > 0$

$\Rightarrow \quad 9a + 3|b| + c > 0$

Also, $0, -2, 2$ are lying between the roots

therefore $c > 0$ and $4a + 2|b| + c > 0$

53.(ABC)

$\sin^2 x - a\sin x + b = 0$ has only one solution in $(0, \pi)$

$\Rightarrow \quad \sin x = 1$ gives one solution and $\sin x = \alpha$ gives other solution such that $\alpha > 1$ or $\alpha \leq 0$

$\Rightarrow \quad (\sin x - 1)(\sin x - \alpha)$ is the same equation as $\sin^2 x - a\sin x + b = 0$

$\Rightarrow \quad 1 + \alpha = a$ and $\alpha = b$

$\Rightarrow \quad 1 + b = a$ and $b > 1$ or $b \leq 0$

$\Rightarrow \quad b \in (-\infty, 0] \cup [1, \infty)$

and $\quad a \in (-\infty, 1] \cup [2, \infty)$

54.(AD)

Section 4	Match Matrix Type

55-57 55.(B) 56.(A) 57.(A)

$$A = \frac{n(n+1)(2n+1)}{6}$$

Say $B = (B' - C)$

Where $B' = \sum_{m=1}^{n} \frac{m(m+1)}{2} = \frac{1}{6} n(n+1)(n+2)$

$$C = \frac{1}{2}\sum_{r=1}^{n} r = \frac{n(n+1)}{4}$$

$\Rightarrow \quad B = (B' - C)$

$$= \frac{1}{6} n(n+1)(n+2) - \frac{1}{4} n(n+1) = \frac{n(n+1)}{12}(2n+1)$$

$\Rightarrow \quad \frac{A}{B} = 2$

$$\left(\frac{b}{c} + \frac{c}{b}\right) + \left(\frac{c}{a} + \frac{a}{c}\right) + \left(\frac{a}{b} + \frac{b}{a}\right) \geq 6; \qquad \text{(by A.M. } \geq GM\text{)}$$

$$\frac{2x^2}{1-x} + \frac{2y^2}{1-y} + \frac{2z^2}{1-z} \geq 1$$

By A.M $\geq$ G.M

$$\left(\frac{x+y+z}{3}\right) \geq (xyz)^{\frac{1}{3}}$$

$$\left(\frac{x+y+z}{3}\right)^3 \geq xyz$$

From given condition $xyz = \left(\frac{x+y+z}{3}\right)$

$\Rightarrow \quad \left(\frac{x+y+z}{3}\right)^2 \geq 1 \quad \Rightarrow \quad x+y+z \geq 3$

58-60 58.(D) 59.(A) 60.(B)

$f(2) = 2^2$, $f(3) = 2^3$

$a = f(3) = 8$, $b = 10$, $c = 12$

$\Delta = \sqrt{15 \times 7 \times 5 \times 3} = 15\sqrt{7}$

$\therefore \quad R = \frac{abc}{4\Delta} = \frac{8 \times 10 \times 12}{4 \times 15\sqrt{7}} = \frac{16}{\sqrt{7}}$

$$r = \frac{\Delta}{s} = \frac{15\sqrt{7}}{15} = \sqrt{7}$$

$\cos A = \frac{3}{4}$

$\therefore \quad \sec^2 A = \frac{16}{9} \quad \Rightarrow \quad \tan^2 A = \frac{7}{9}$

Section 5	Single Integer Answer Type

61.(7) $a_n = \dfrac{(n+2)}{n(n+1)(n+3)} = \dfrac{(n+2)^2}{n(n+1)(n+2)(n+3)}$

$$= \frac{A}{(n+2)(n+3)} + \frac{B}{(n+1)(n+2)(n+3)} + \frac{C}{n(n+1)(n+2)(n+3)} \text{ (say)}$$

$\Rightarrow \quad (n+2)^2 = An(n+1) + Bn + C$

$\Rightarrow \quad n^2 + 4 + 4n = An^3 + An + Bn + C$

Now comparing the coefficient of powers of n,

$\Rightarrow \quad A = 1 \qquad A + B = 4 \qquad C = 4 \qquad B = 3$

So, $\quad a_n = \dfrac{1}{(n+2)(n+3)} + \dfrac{3}{(n+1)(n+2)(n+3)} + \dfrac{4}{n(n+1)(n+2)(n+3)}$

So, $\quad S_n = \displaystyle\sum_1^n a_n = \sum_1^n \left[\frac{1}{n+2} - \frac{1}{n+3}\right] + \frac{3}{2}\sum_1^n \left[\frac{1}{(n+1)(n+2)} - \frac{1}{(n+2)(n+3)}\right]$

$$+ \frac{4}{3}\sum_1^n \left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)}\right]$$

$$= \left[\frac{1}{3} - \frac{1}{n+3}\right] + \frac{3}{2}\left[\frac{1}{6} - \frac{1}{(n+2)(n+3)}\right] + \frac{4}{3}\left[\frac{1}{6} - \frac{1}{(n+1)(n+2)(n+3)}\right]$$

$$= \frac{29}{36} - \frac{1}{n+3} - \frac{3}{2(n+2)(n+3)} - \frac{4}{3(n+1)(n+2)(n+3)}$$

62.(1) $S = \displaystyle\sum_{a=1}^{1999} \sqrt{1 + \frac{1}{a^2} + \frac{1}{(a+1)^2}} = \sum_{a=1}^{1999} \sqrt{\frac{a^4 + 2a^3 + 3a^2 + 2a + 1}{a^2(a+1)^2}}$

$$= \sum_{a=1}^{1999} \frac{a^2 + a + 1}{a(a+1)} = \sum_{a=1}^{1999} \left(1 + \frac{1}{a(a+1)}\right) = 1999 + \sum_{a=1}^{1999} \left(\frac{1}{a} - \frac{1}{(a+1)}\right)$$

$$= 1999 + \left(1 - \frac{1}{2000}\right) = 1999 + \frac{1999}{2000}$$

Hence $p = 1999$ and $q = 2000$

63.(1) Given $a\alpha^2 + b\alpha + 1 = 0$ .... (1)

Also $\bar{a}\,\bar{\alpha}^2 + \bar{b}\,\bar{\alpha} + 1 = 0 \quad \Rightarrow \quad \dfrac{\bar{a}}{\alpha^2} + \dfrac{\bar{b}}{\alpha} + 1 = 0$ (as $|\alpha| = 1$)

$\alpha^2 + \bar{b}\alpha + \bar{a} = 0$ ……..(2)

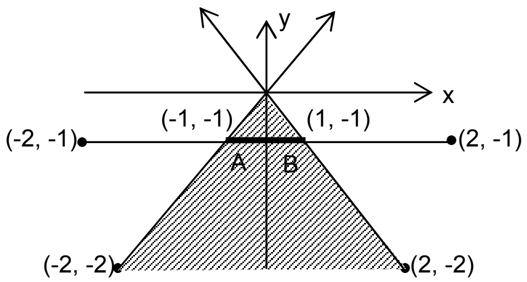
From (1) and (2)

$$\frac{\alpha^2}{\bar{a}b - \bar{b}} = \frac{\alpha}{1 - |a|^2} = \frac{1}{a\bar{b} - b}$$

$\Rightarrow \quad \dfrac{\bar{a}b - \bar{b}}{1 - |a|^2} = \dfrac{1 - |a|^2}{a\bar{b} - b}$

$\Rightarrow \quad |a\bar{b} - b| = 1 - |a|^2 = \dfrac{3}{4}$

64.(2) From figure it is clear that required length of line segment is $AB = 2$



65.(2) $\lim_{n\to\infty}\sum_{r=0}^{n}\frac{{}^nC_r}{n^r(r+3)} = \lim_{n\to\infty}\sum_{r=0}^{n}\frac{{}^nC_r}{n^r}.\int_0^1 x^{r+2}dx.$

$$= \int_0^1\left(x^2\lim_{x\to\infty}\sum_{r=0}^{n}{}^nC_r\left(\frac{x}{n}\right)^r\right)dx = \int_0^1\left(x^2\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n\right)dx = \int_0^1 x^2.e^x.dx = e-2$$

But $\quad e-2 = e-x \quad \therefore \quad x = 2$

66.(3) We have $\lim_{x\to 0}\left[\sum_{r=0}^{n}(-1)^r\ {}^nC_r\left(\sum_{k=0}^{n-r}{}^{n-r}C_k.x^k.2^k\right)\left(x^2-x\right)^r\right]^{1/x}$

$\Rightarrow \quad \lim_{x\to 0}\left[\sum_{r=0}^{n}(-1)^r\ {}^nC_r\,(1+2x)^{n-r}\left(x^2-x\right)^r\right]^{1/x}$

$\Rightarrow \quad \lim_{x\to 0}\left[\left(1+2x-x^2+x\right)^n\right]^{1/x} \Rightarrow \lim_{x\to 0}\left(1+3x-x^2\right)^{n/x}$

$\Rightarrow \quad e^{\lim_{x\to 0}\frac{\left(3x-x^2\right).n}{x}} \Rightarrow e^{3n}\left[\because \lim_{x\to 0}\left[1+f(x)\right]^{g(x)} = e^{\lim_{x\to 0}f(x).g(x)}\right]$

$\Rightarrow \quad e^{3n}$

$\because \quad \lambda = 3$

67.(2) Here, $S = \sum_{k=1}^{\infty}\frac{1}{K^2} = 1^2+\left(\frac{1}{2}\right)^2+\left(\frac{1}{3}\right)^2+\ldots\ldots.+$

$$t = \sum_{k=1}^{\infty}\frac{(-1)^{K+1}}{K^2} = (1)^2-\left(\frac{1}{2}\right)^2+\left(\frac{1}{3}\right)^2-\ldots\ldots\ldots\ldots\ldots\ldots$$

$$S-t = 2.\left\{\left(\frac{1}{2}\right)^2+\left(\frac{1}{4}\right)^2+\left(\frac{1}{6}\right)^2+\ldots\ldots..+\right\} = 2.\frac{S}{4} = \frac{S}{2}$$

$\Rightarrow \quad \frac{S}{2} = t \quad \therefore \quad \frac{S}{t} = 2$

68.(2) Number 777^{777} is divisible by 7 so $555^{555}+666^{666}+888^{888}+999^{999}$ should be written as

$\left(999^{999}+555^{555}\right)+\left(888^{888}+666^{666}\right)$

$=\left[999^{999}+2^{999}\right]+\left[555^{555}-2^{555}\right]-\left[2^{999}-2^{555}\right]+\left[888^{888}-6^{888}\right]+\left[666^{666}+6^{666}\right]+\left[6^{888}-6^{666}\right]$

$=\left(999^{999}+2^{999}\right)$ is divisible by 1001 and 1001 is divisible by 7.

Hence it is divisible by 7.

$888^{888}-6^{888}$ is divisible by 882 and 882 is divisible by 7. Hence it is divisible by 7.

$2^{999}-2^{555}=2^{555}\left(2^{444}-1\right)=2^{555}\left(2^{3\times148}-1\right)=2^{555}\left(8^{148}-1\right)$. Hence it is divisible by 7,

$6^{888}-6^{666}=6^{666}\left[6^{222}-1\right]=6^{666}\left[36^{111}-1\right]$. Hence it is divisible by 35. So it is also divisible by 7.

$666^{666}+6^{666}=(665+1)^{666}+(7-1)^{666}$ is divided by 7 we get remainder $=1^{666}+1^{666}=2$.

Hence the remainder of number divided by 7 is 2.

69.(9) All the possible number are ${}^9C_5$ (none containing the digit 0)

= 126.

Total numbers starting with

$1={}^8C_4=70$

1				

(using 2,3,4,5,6,7,8,9) Total number starting with $23={}^6C_3=20$

2	3			

(using 4,5,6,7,8,9) Total number starting with $245={}^4C_2=6$

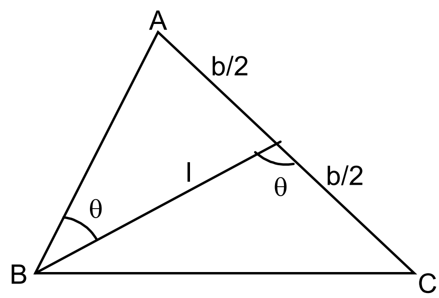
2	4	5		

(using 6,7,8,9)

97^{th} number =

2	4	6	7	8

70.(4) Let ABC be the triangle with D as the midpoint of AC. $BD=l$ (given)



Area of $\Delta ABC=\frac{b^2}{2}\sin A$.

Applying cosine rule to ΔABD, we get $\frac{b^2+\frac{b^2}{4}-l^2}{b^2}=\cos A$

$$\Rightarrow \quad b^2=\frac{4l^2}{5-4\cos A}$$

Hence $\Delta=2l^2\frac{\sin A}{5-4\cos A}$

$$\Rightarrow \quad \Delta=2l^2\frac{\frac{2t}{1+t^2}}{5-\frac{4\left(1-t^2\right)}{1+t^2}} \qquad \left[\text{where } t=\tan\frac{A}{2}>0\right]$$

$$\Rightarrow \quad \Delta=\frac{4l^2t}{9t^2+1}=\frac{4l^2}{9t+\frac{1}{t}}$$

$$\Rightarrow \quad \Delta\le\frac{2l^2}{3} \qquad \left[\text{Since} \quad 9t+\frac{1}{t}\ge2\cdot\left(\frac{1}{t}\cdot9t\right)^{\frac{1}{2}}=6\right]$$

Hence maximum area $=\frac{2l^2}{3}$

Equality holds when $9t=\frac{1}{t} \quad\Rightarrow\quad t=\frac{1}{3}$ [since $t>0$] $\quad\Rightarrow\quad \cos A=\frac{4}{5}$

71.(6) $S_n = \sum_{r=1}^{n} \frac{r^2+1}{r(r+1)} 2^{r-1} = \sum_{r=1}^{n} \left(\frac{2r}{r+1} - \frac{r-1}{r} \right) 2^{r-1} = \frac{n}{n+1} \cdot 2^n$

72.(4) $b_n = a^n r^{n\frac{(n-1)}{2}} \Rightarrow \sum_{n=1}^{\infty} ar^{\frac{(n-1)}{2}} = \frac{a}{1-\sqrt{r}} = 32$

73.(1) The roots of the equation

$x = \frac{-n \pm \sqrt{n^2 - 4n}}{2}$ may be integers if $n^2 - 4n = I^2$ where I is an integer.

$\Rightarrow \quad n^2 - 4n = I^2 = 0 \quad$ ....(1)

$\Rightarrow \quad n = 2 \pm \sqrt{4 + I^2}$

Now n is an integer $\Rightarrow \sqrt{4+I^2}$ should be an integer.

$\Rightarrow \quad 4 + I^2 = k^2$ where k is an integer.

$\Rightarrow \quad k^2 - I^2 = 4$

which is possible only when $k = \pm 2, I = 0.$

putting $I = 0$ in (1),

$n^2 - 4n = 0 \quad \Rightarrow \quad n = 0, 4.$ For both these values,

$x^2 + nx + n = 0$ has integral roots.

$\therefore \quad n = 0, 4 \quad \Rightarrow \quad n^2 - 4n = 0$

74.(2) Let $f(x) = x + ax^{-2}$

$f'(x) = 1 - 2ax^{-3} = 0 \quad \Rightarrow \quad x = (2a)^{1/3}$

$f''(x) = 6ax^{-4} > 0 \ \forall \ x \in (0, \infty)$ (as 'a' is a natural number)

so $x = (2a)^{1/3}$ is a point of global minima

Thus $(2a)^{1/3} + a(2a)^{-2/3} > 2$

$\Rightarrow \quad a > \frac{32}{27} \quad \Rightarrow \quad a \geq 2.$ As 'a' is a natural number.

Alternative solution

$x + ax^{-2} > 2 \quad \Rightarrow \quad x^3 - 2x^2 + a > 0$

Let $f(x) = x^3 - 2x^2 + a$

Since $f(x) > 0 \ \forall \ x \in (0, \infty), f(0) > 0$ and min $f(x) > 0$

$\Rightarrow \quad a > 0$ and for minimum $f(x)$

$f'(x) = 3x^2 - 4x \quad \Rightarrow \quad x = 0, 4/3$

$f(4/3) > 0 \quad \Rightarrow \quad a > \frac{32}{27}.$

75.(9) Let roots of $x^3 - ax^2 + bx - c = 0$ be α, β, γ

$\Rightarrow \quad a = \alpha + \beta + \gamma, b = \alpha\beta + \beta\gamma + \gamma\alpha, c = \alpha\beta\gamma$

Now, $(\alpha+1), (\beta+1), (\gamma+1)$ are the roots of $x^3 + px^2 + qx - 9 = 0$

$\Rightarrow \quad (\alpha+1)(\beta+1)(\gamma+1) = 9$

$\Rightarrow \quad \alpha\beta\gamma + \alpha + \beta + \gamma + \alpha\beta + \beta\gamma + \gamma\alpha + 1 = 9 \quad \Rightarrow \quad a + b + c = 8$